

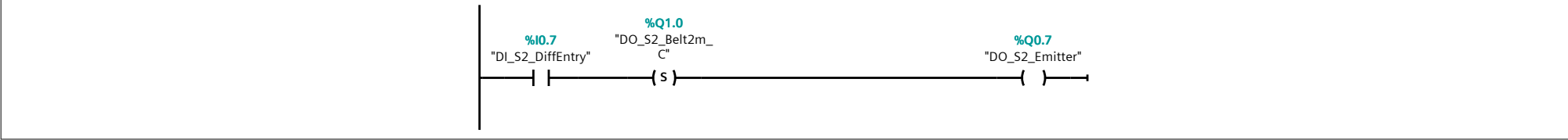
Station2 [FC2]

Station2 Properties							
General							
Name	Station2	Number	2	Type	FC	Language	LAD
Numbering	Automatic						
Information							
Title	Station 2 — Assembly	Author		Comment	Station 2 (Assembly). -The Two-Axis Pick & Place assembles a Product Lid on-to a Product Base. -A diffuse sensor at the entry (DI_S2_DiffEntry) starts belt C (DO_S2_Belt2m_C); once the part is in position the stop blade (DO_S2_StopBlade0) holds it. -The arm executes a fixed pick-and-place sequence by driving the Z axis first (r_S2_CmdMoveZ = 10.0 to lower, 0.0 to raise), then the X axis (r_S2_CmdMoveX = 10.0 to move over the base, 0.0 to return). -The gripper is closed (DO_S2_Grabe) while the arm is above the base. - Rotation (DO_S2_Rotate-Right / DO_S2_RotateLeft) is used to place the lid with the correct orientation. -The cycle ends when the arm returns to the home sensor (DI_S2_DiffArmHome).	Family	
Version	0.1	User-defined ID	Youness Tebbaq				

Name	Data type	Default value	Comment
Input			
Output			
InOut			
Temp			
Constant			
▼ Return			
Station2	Void		

Network 1: Start belt C and latch the emitter when an item is present

Latch the emitter (DO_S2_Emitter) when a part enters station 2 (DI_S2_DiffEntry). The emitter stays ON because the network ORs the latched output back into itself. The emitter is cleared in Network 4 by the stop blade 1 sensor.



Network 2: Extend stop blade 0 when a part is at zone 1

When a part reaches the diffuse sensor for stop blade 0 (DI_S2_DiffStop0), extend the stop blade (DO_S2_StopBlade0) to hold the part in place for assembly.



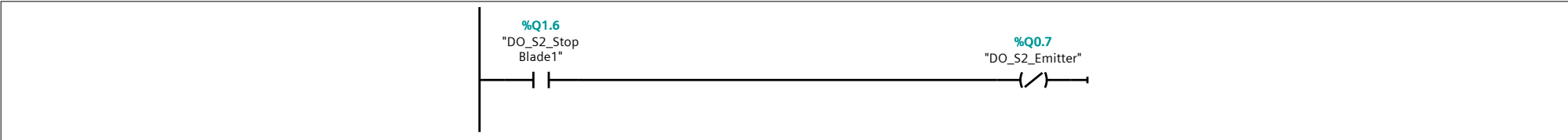
Network 3: Extend stop blade 1 when a part is at zone 2

When a part reaches the diffuse sensor for stop blade 1 (DI_S2_DiffStop1), extend the stop blade (DO_S2_StopBlade1) to hold the second part in place for assembly.

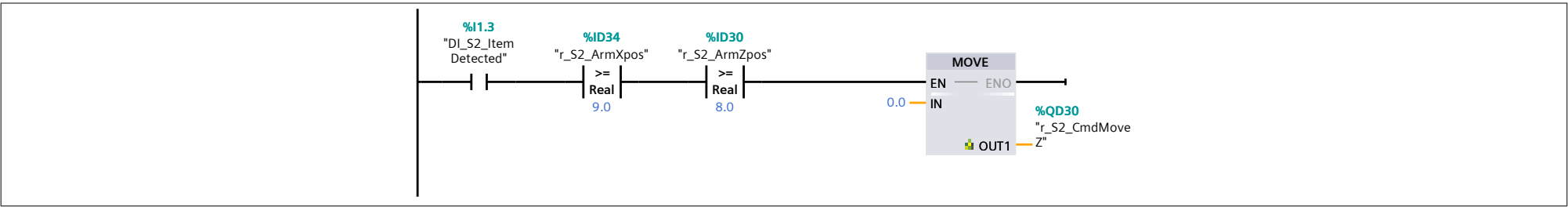


Network 4: Clear the emitter when stop blade 1 is extended

"When stop blade 1 is extended (DO_S2_StopBlade1), the part is in position, so stop emitting. (Rising edge of stop blade 1 resets the emitter flag.)



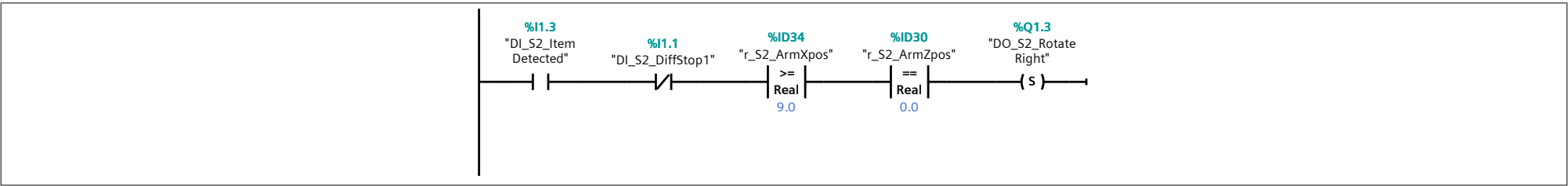
Totally Integrated Automation Portal		
Network 5: Clear the station 1 emitter when stop blade 0 is extended When stop blade 0 is extended (DO_S2_StopBlade0), upstream station 1 is signaled to stop emitting. This back-pressure prevents part accumulation between stations.		
Network 6: Step 1 — Lower the arm to pick position Step 1 of the pick-and-place sequence. Lower the Z axis to the pick position (r_S2_CmdMoveZ = 10.0) when: - an item is detected (NOT DI_S2_ItemDetected — meaning the part HAS left the entry diffuse sensor and is in the assembly area), - the arm is not rotating (NOT DO_S2_RotateRight, NOT DO_S2_RotateLeft), - a part is in zone 1 (DI_S2_DiffStop0), AND - the X axis is at the home position (r_S2_ArmXpos = 0.0). After driving Z down, the gripper is closed (SET DO_S2_Grabe).		
Network 7: Step 2 — Raise the arm back to the top Step 2: raise the Z axis back to the top (r_S2_CmdMoveZ = 0.0) when: - an item is detected (DI_S2_ItemDetected), - the Z axis is in the pick position (r_S2_ArmZpos >= 9.0) The arm lifts the lid off its pickup point to brings it over the base.		
Network 8: Step 3 — Drive the X axis over the base Step 3: move the arm horizontally to the base position (r_S2_CmdMoveX = 10.0) when: - an item is detected (DI_S2_ItemDetected), - Z is at the home (top) position (r_S2_ArmZpos = 0.0), - the arm is not rotating (NOT DI_S2_ArmRotating), AND - the arm is not currently rotating right (NOT DO_S2_RotateRight).		
Network 9: Step 4 — Lower the arm onto the base Step 4: lower the Z axis onto the base position (r_S2_CmdMoveZ = 10.0) when: - an item is detected (DI_S2_ItemDetected), - stop blade 1 is set (DI_S2_DiffStop1), - the X axis has reached the base position (r_S2_ArmXpos >= 9.0), AND - Z is at the home (top) position (r_S2_ArmZpos = 0.0).		
Network 10: Step 5 — Raise the arm back to the top Step 4: raise the Z axis back to the top (r_S2_CmdMoveZ = 0.0) when: - an item is detected (DI_S2_ItemDetected), - the X axis is over the base (r_S2_ArmXpos >= 9.0), AND - Z has reached the base position (r_S2_ArmZpos = 8.0). The arm lifts the Final Product(lid+base) off its pickup point .		



Network 11: Step 6 — Start the rotation

Step 6: energise the rotate-right coil (SET DO_S2_RotateRight) when:

- an item is detected (DI_S2_ItemDetected),
- stop blade 1 is clear (NOT DI_S2_DiffStop1),
- the X axis is over the base belt (r_S2_ArmXpos >= 9.0), AND
- Z is at the top (r_S2_ArmZpos = 0.0).

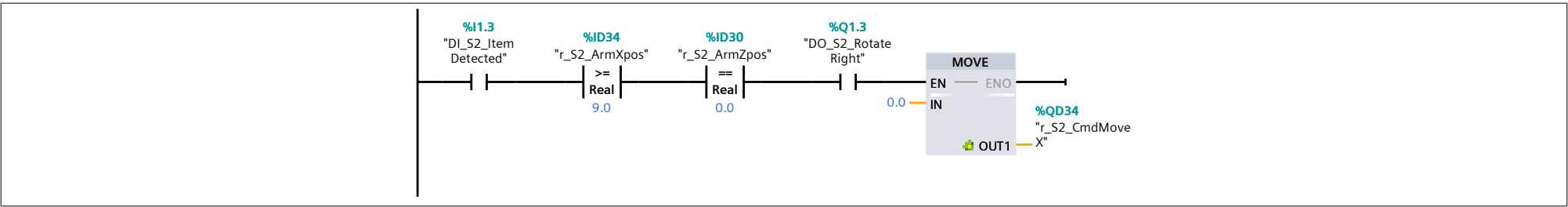


Network 12: Step 7 — Return the X axis to home

Step 7: drive the X axis back to the home position ($r_{S2_CmdMoveX} = 0.0$) when:

- an item is detected (DI_S2_ItemDetected),
- Z is at the home position (r_S2_ArmZpos = 0.0),
- X is at the top (r_S2_ArmXpos >= 9.0), AND
- the rotate-right coil is still energised (DO_S2_RotateRight).

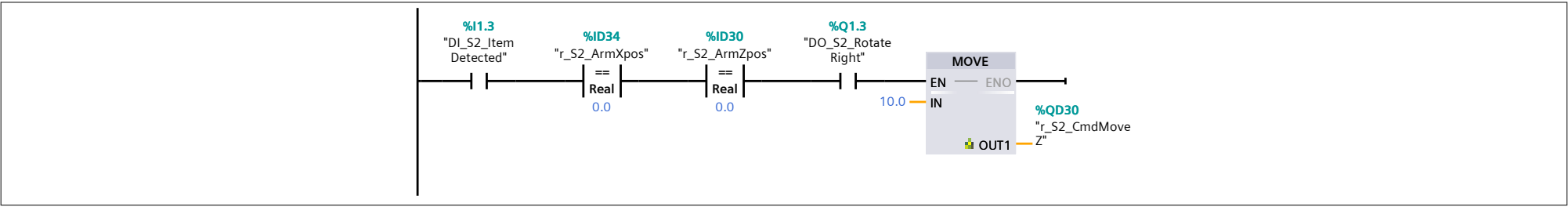
The arm returns while continuing to rotate.



Network 13: Step 8 — Lower the arm onto the outfeed belt

Step 8: lower the Z axis onto the outfeed belt position (r_S2_CmdMoveZ = 10.0) when:

- an item is detected (DI_S2_ItemDetected),
- the X axis is at the home position (r_S2_ArmXpos = 0.0), AND
- Z is at the home (top) position (r_S2_ArmZpos = 0.0).

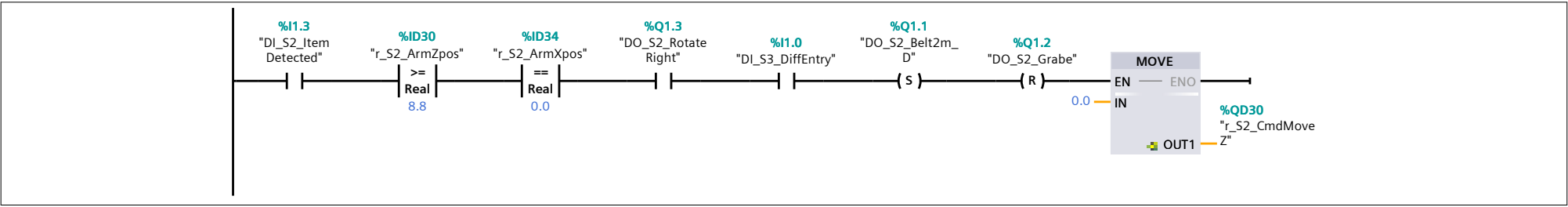


Network 14: Step 9 — Release the part onto the outfeed belt

Step 9: set the outfeed belt (DO_S2_Belt2m_D) release the gripper (RESET DO_S2_Grabe) and drive Z to zero when:

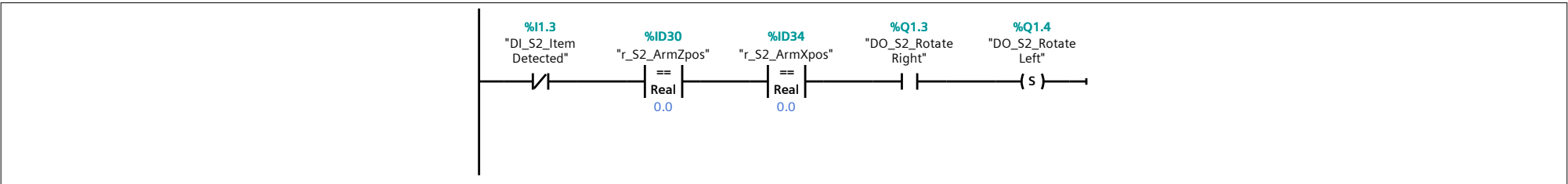
- an item is detected (DI_S2_ItemDetected),
- Z is high enough to clear the base (r_S2_ArmZpos >= 8.8),
- X is at the home position (r_S2_ArmXpos = 0.0),
- rotate-right is still energised (DO_S2_RotateRight), AND
- the FP is present ("DI_S3_DiffEntry") at the outfeed belt .

The assembled part drops onto the outfeed belt and leaves the station.



Network 15: Idle — Set rotate-left when the FP is released onto the outfeed belt

Idle state: when the FP is released onto the outfeed belt (NOT DI_S2_ItemDetected) and rotate coil is still energised, set rotate-left (DO_S2_RotateLeft). This counter-rotates the gripper back to its home orientation so it is ready for the next cycle.



Network 16: Home position — clear both rotation coils

When the arm reaches the home sensor (DI_S2_DiffArmHome), Z and X are both at zero, and both rotation coils still energised.
The arm is now in its home position and ready to start a new cycle.

